

Mengxiao Wang

Texas — jacksonwang@tamu.edu — 6128145421 — jacksonwang.site — github.com/jackson-cmd

Research Focus: Specialized in targeted attacks and defenses for domain-specific LLM agents, with emphasis on financial trading systems. Research covers prompt injection, jailbreak, and prompt leaking attacks against LLM-based autonomous agents in high-stakes domains.

Education

Texas A&M University Ph.D. in Computer Science (<i>Advisor: Dr. Nitesh Saxena</i>) <i>GPA: 4.0/4.0</i>	College Station, TX Aug 2023 – Present
Carnegie Mellon University Master of Science in Electrical and Computer Engineering <i>GPA: 3.78/4.0</i>	Pittsburgh, PA Sep 2021 – May 2023
University of Minnesota, Twin Cities Bachelor of Arts in Computer Science, Minor in Statistics <i>GPA: 3.815/4.00</i>	Minneapolis, MN Jan 2018 – May 2021

Skills

Languages: C/C++, Python, Go, Java, JavaScript/TypeScript, X86 Assembly, SQL, Rust
Technologies: Docker, Linux, MySQL, Git, Kali Linux (Burp Suite, Nmap, Metasploit), Design Patterns

Work Experience

Inspur <i>LLM Security Research Intern</i>	Jinan, China May 2024 – July 2024
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- Assisted in developing a large language model (LLM) using LangChain and the ChatGLM model with API implementations. Utilized Python, FastAPI, and Flask to create a local deployment system for clients, enabling document analysis and summarization. The client could select different models to adjust LLM autonomy, with parameters like temperature and max tokens being configurable.
- Developed a fuzzing tool to test LLM vulnerabilities (e.g., prompt injection, unauthorized access), generating 1,000+ test cases. Discovered three novel methods to bypass content restrictions: context manipulation through subtle prompt phrasing, recursive question embedding, and hidden token injection via encoding tricks.
- Implemented an initial input-output filter with a high false positive rate (22.3%), which was reduced to 4% after integrating semantic analysis. Also applied strict file permissions and access control to mitigate LLM unauthorized file access.

Carnegie Mellon University <i>Research Assistant (CyLab)</i>	Pittsburgh, PA July 2021 – May 2023
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- Led the effort to analyze TypeScript code flows using taint analysis, contributing to a crawl of over 2.1 million npm packages and improving the tool's detection of command injection vulnerabilities by 40%.
- Helped the team submit 30+ new CVEs related to code execution and command injection vulnerabilities in npm packages by refining detection mechanisms and enhancing vulnerability discovery.
- Manually crafted complex input flows, identifying vulnerabilities missed by automated tools, and improved the tool to handle edge cases such as type-specific inputs, further enhancing its detection capabilities.

Inspur <i>Full Stack Intern</i>	Jinan, China May 2018 – August 2018
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- Scraped province-wide maps via Amap API with Python, integrating external data to boost coverage by 50% and improve MongoDB accuracy. Built the frontend in React and backend in Java (Spring

Boot), reducing map query load times by 25%.

- Performed security tests with Burp Suite Pro, fixing SQL injection and XSS. Developed proof-of-concept exploits using Python and Bash, reducing attack surface by 15% and exploit risk by 30%.

Publications

SoK: Trading Agents or Market Crashers? Dissecting Robustness and Security Failures in Academic Financial LLM Trading Schemes M Wang, N Saxena. Under review, 2026

AgentNets as Market Attackers: From Information Injection to Financial Microstructure Manipulation M Wang, N Saxena. Under review, 2026

Demystifying Progressive Web Application Permission Systems Wang, M., & Gu, G. In Proc. of the 56th IEEE/IFIP International Conference on Dependable Systems and Networks (DSN), 2026.

Cutting through the Confusion: A Measurement Study of Homograph Domains in Ethereum Name Service J Huang, SR Chintapalli, M Wang, G Gu. In Proceedings of The Web Conference (WWW), 2025

Application of Computer Vision in Logistics Warehouse Safety Management Wang, M. Software, 41(10), 180-183. ISSN:1003-6970

Awards & Grants

OpenAI Researcher Access Program Grant 2025

Professional Service

- **Program Committee**, ACM The Web Conference (WWW), 2025
- **External Reviewer**, Network and Distributed System Security Symposium (NDSS), 2023, 2024
- **External Reviewer**, USENIX Security Symposium (USENIX Security), 2024, 2025
- **External Reviewer**, IEEE Symposium on Security and Privacy (IEEE S&P), 2023
- **External Reviewer**, ACM Conference on Computer and Communications Security (CCS), 2023–2025

Student Supervising/Mentoring

Landry Taylor [Bachelor of Business Administration] Spring 2026 - Present
Texas A&M University, College Station, TX, USA